- **Employed and self-employed**. An increasing portion of workers are self-employed, more than working under someone (employed).
- **Full-time and Part-time**. Increasing part-time jobs due to people having other responsibilities, increased leisure time activities or inability to find full-time employment.
- **Employment in the Formal Economy and Informal Economy**. A formal economy is protected through worker legislation and leads to **secure employment** whereas an informal economy has no trade union/government representation and often gives **insecure employment**.
- **Private and Public Sector Employment**. Public sector employment provides high wages and non-monetary benefits (high-quality employment), but private sector employment is rising due to privatisation.
- It is important to note that it is not easy to compare countries in unemployment patterns; countries not only have differing economic conditions and labour markets but have different cultural systems, political agendas, and government objectives, which can result in effective comparisons.

- **Patterns and Trends of Unemployment**:
  - **Frictional Unemployment**: unemployment is temporary and arises when people are between jobs.
    - **Casual and Seasonal Unemployment** is a form of frictional unemployment.
  - **Structural Unemployment**: unemployment caused by the changing structure of economic activities
    - **Technological, regional and international unemployment**.
  - **Cyclical Unemployment**: unemployment that results from a lack of aggregated demand.
    - Firms could cut wages, but demand-deficient unemployment will exist if workers resist.

3.10. Mobility of Labour

- **Occupational Mobility**: the ability of workers to move from one occupation to another occupation.
- **Factors Affecting Occupational Mobility**:
  - **Quality of education and training**
  - **Information available** on job opportunities and occupation demands.
  - **Barriers to entry and exit**: restrictions in the supply of occupation by trade unions/professional bodies to raise wage rate or Long-term contracts that bind workers.
  - **Time**: opportunities arise over time that may help workers build their qualifications.
- **Geographical Mobility**: the ability of workers to move from one location to another.
- **Factors Affecting Geographical Mobility**:
  - The price and availability of housing
  - Information available on job vacancies
  - Personal life may make it harder for workers to separate.
  - Immigration controls
  - Language barriers
  - Cultural differences
  - Differences in wages and living expenses

3.11. Policies to Reduce Unemployment

- The government will want to increase aggregate demand to increase the need for employment. This helps with cyclical unemployment.
- **Expansionary Fiscal Policy** tools:
  - **Reduce direct or indirect tax** to increase consumer expenditure.
  - **Cut in corporation taxes** to stimulate investment.
  - Increase in **government spending**.
- **Monetary Policy** tools:
  - **Reduce the rate of interest** to increase consumer expenditure and investment.
  - Increasing **money supply**
  - **Lowering exchange rate** to increase net exports.
- **Effectiveness of these policies**:
  - **Firm/consumer optimism** is needed for an effect to take place.
  - **The multiplier effect** must be kept in mind not to be underestimated and cause inflation, which worsens unemployment.
  - The **time lags** of these policies may make them **inefficient or cause adverse effects**. The government may realise the issue late, so when it comes to applying policies, the economy has started to shift upwards, and aggregate demand is increasing again.
- **Supply-side policies** are used to **reduce frictional and structural unemployment** through increasing labour supply:
  - **Increase information on the job market**
  - **Reduce unemployment benefits and income tax rate**
  - **Improved education and training** increase occupational mobility.
  - **Reducing trade union influence**
- **Effectiveness of supply-side policies**:
  - Some of these policies **take a long time to take effect**. e.g. improved education causes an increase in qualifications in the long run, but the skills learned may be outdated.
  - **Response to these policies is unexpected**. e.g. a cut in income tax may not be the incentive individuals need to join the workforce. This is especially the case if it turns out the cause of unemployment has been a lack of demand, which may lead to worsening unemployment.

3.12. Money and Banking

- **Money**: anything that is generally acceptable as a means of payment.
- **~~Functions of Money~~**:
  - **A medium of exchange**

- A **unit of account**. This function is **relevant for future and current transactions** and allows **different values to be added, measured, compared and recorded**.
- A **standard for deferred payment** (advance payment).
- A **store of wealth**, a measure of value over time.
- If any of these **functions break down, money loses its value**, leading to economic collapse.

- **Characteristics of Money**:
  - Generally acceptable
  - Recognisable
  - Portable
  - Divisible
  - Homogeneous (identical)
  - Limited in supply
  - Not easy to counterfeit
- **Money Supply**: consists of circulation + relevant deposit.
- Measuring money supply is difficult, as definitions and functions of money change and vary over time. Main two methods:
  - **Narrow Money**: money that can be spent directly. Used as a medium of exchange, sometimes referred to as Monterey base.
  - **Broad Money**: money used for serving and spending.

The Quantity Theory of Money

- **The Quantity Theory of Money**: the theory that links inflation in an economy to changes in the money supply.
  - **Fisher equation**: the statement/identity of MV=PY.
    - M=money supply, V=velocity of circulation (number of times money is passed down). MV gives us nominal GDP, which is consumer-centred.
    - P=price level, Y=real GDP (output of the economy). PY gives us nominal GDP, firm-centred.
    - **What consumers buy should match what firms sell**; hence, MV=PY. They equal the total expenditure of the economy.
  - This equation relates to the theory by keeping V and P constant (not affected by a change in the money supply) so that a change in the money supply causes an equal percentage change in the price level.
  - For example, the money supply may initially be US$80 billion, the velocity of circulation 5, price level 100, and output 4 billion. If *V* and *Y* are unchanged, an increase in the money supply by 50% to 120 would also cause the price level to rise by 50% to 150.
  - The monetarist view is that a change in money supply causes inflation. At the same time, Keynesians argue that the equation cannot be turned into a theory since *V* and *Y* can change with a change in the money supply, so no predictions can be made about what effect a change in *M* will have on *P*.

Functions of Commerical Banks

- **Commercial Banks**: a financial institution whose purpose is to accept deposits from people and provide loans and other facilities.

- **Provide Deposit Accounts** for their customers:
  - **Demand deposit account**: provides easy and quick access to the money in the account. They are used for **receiving and making payments**.
  - **Saving deposit account**: used mainly as a way of saving.
- **Hold a range of assets**, such as:
  - Cash
  - Government securities - bills and bonds issued by the government to raise money.
  - Equities (Capital and assets for firms).
- Most of their profit is made through **lending**.
  - **Overdraft**: the opportunity for customers to overdraw from their deposit account by a certain value. **Interest rate is charged** (that's how profit is made).
  - **Loans** can be agreed upon for a particular purpose and a set amount of time; interest **is charged** on the loan's value and will have to be paid regardless of whether the loan has been used or not.
- **Reserve Ratio**: the **proportion of liquid assets to total deposits** that commercial banks keep.
- **The Bank Credit Multiplier** is the process by which banks can make more loans than deposits.
  - The bank credit multiplier (after loans have been made):
    - $\frac{\text{Value of new assets created}}{\text{Value of change in liquid assets}}$
  - The bank credit multiplier (in advance):
    - $\frac{100}{\text{liquidity ratio}}$
  - For example, if the bank credit multiplier is ten and liquid assets rise by 25m dollars, total deposits can increase by 10x25=250m dollars.
  - Banks don't always lend as much as the bank credit multiplier implies it can due to a lack of borrowers or creditworthy borrowers.
  - Change in reserve ratio may happen due to a change in the **proportion of peoples' deposits**, a change in **other banks' lending policies**, or as a **requirement from the central bank**.
  - Central banks influence Commercial banks through the selling of government securities.
    - To reduce bank loans, they sell government securities, causing purchasers to draw from their deposits in the Commercial bank, reducing liquid assets.
- **Capital Ratio**: a commercial bank's available **financial capital as a percentage of its riskier assets**.
  - Financial capital is retained profits, newly issued shares, and sometimes, government securities (if the economy gets into difficulties).
  - For example, if the ratio is 7%, that would mean that 7% of its risk assets are covered by readily available financial capital.
  - The higher the ratio, the more unexpected losses it can experience without going out of business.
  - It is designed to protect bank customers during a financial crisis and to promote stability in the banking sector by discouraging excessive risk-taking.

- **The Objectives of Commercial Banks:**
  - **Profitability**: achieve high profits for its shareholders by lending. Conflicts with the other two objectives
  - **Liquidity**: enough liquid assets (cash and short-term securities that will soon be turned into cash) to meet the expected request by their customers to withdraw their money in cash. However, it is not very profitable (since they hold cash as a reserve).
  - **Security**: keeping their customers by making sure they are financially well, through having sufficient financial capital to cover their risker loans.

Causes and Changes in the Money Supply

- **Increase in Commerical Bank Lending**
  - Commercial banks **create money through lending**.
  - But have to keep in **mind the liquidity ratio** (the proportion of liquid assets to total liabilities); the **lower the proportion of liquid assets** commercial banks keep, the **more they can lend**, but you have to be careful not to keep it **too low** or else they risk **losing customers**.
- An **increase in government spending** financed by **borrowing from the central bank**
  - The role of a central bank is to be a **bank for Commercial banks** (keep deposits and lend to), act as the **banker to the government** and **implement the government's monetary policy**.
  - When the government experiences a **budget deficit**, it **draws money** from its account, **increasing deposits** and **liquid assets**.
- An **increase in government spending** was financed by **borrowing from the Commercial bank**
  - The government has accounts in both central and Commercial banks.
  - Commercial banks buying **short-term government securities** (counted as liquid assets, the basis for loans) issued by the government increases their **lending power**.
- **The sale of government bonds to private-sector financial institutions**
  - This occurs when the **interest rate is low, and central banks want to increase aggregate demand**.
  - **Quantitive easing**: central bank buying government bonds from the private sector to increase money supply.
  - In return for buying, the central bank **credits the account** of Commercial banks, leading to **more assets and lending**. This increases **money supply and reduces short- and long-term interest rates, boosting investment and consumer expenditure**, therefore increasing **aggregate demand**.
- **More money entering than leaving the country.**
  - **Total Currency Flow**: the current + capital + financial balances of the balance of payments.
  - It is also the Total Net inflow or outflow of money from international transactions.
  - An inflow into the economy, for example, export revenue, exceeding the outflow, for example, import expenditure, will increase the money supply.

3.13. Policies to Reduce Inflation

- **Demand-Pull Inflation**: a rise in prices due to exceeding aggregate demand. There are not enough resources/supply to meet these demands. Hence a price increase.
- **Cost-Push Inflation**: arising when the economy is operating at less than full employment. This creates high costs, hence an increase in prices.
- In practice, It's difficult to tell between the two since a combination usually causes inflation.

Policies to Reduce Demand-Pull Inflation

- **Contractionary Fiscal Policy**: policies to reduce aggregate demand by reducing government spending and/or increasing tax.
- **Contractionary Monetary Policy**: policies to decrease aggregate demand by increasing interest rates, dwindling money supply (using credit control) and increasing foreign exchange rates.
- **Supply-side Policies** are mainly used for cost-push inflation but can be used here to **stimulate supply to meet the exceeding demand** and bring the market back to equilibrium.

The effectiveness of these policies to reduce demand-pull inflation:

- **Higher income tax** may result in a **raised cost of production for firms**, as employees may demand higher wages or leave the labour force altogether.
- **Interest rate** increases may **not result in discouraging consumer expenditure. There is no guarantee that commercial banks will increase interest rates with the central bank's growth, and** consumers may **not reduce spending** if they are optimistic about the future.
- **Increased interest rates** may have **adverse effects on investment**. The cost of borrowing will be high, and the opportunity cost of using retained profit to invest will be high. If investment falls below depreciation, capital stock declines, aggregate supply drops, and prices may rise.
- **Supply-side policies** in the **short run may cause inflation. Increased government spending on education and training**, and cuts **in income tax, causing** an **increase in aggregate demand**.

Policies to Reduce Cost Push Inflation

- **In short**, the central bank may **raise the exchange rate**, resulting in reduced costs pressuring domestic firms to cut **costs**.
- **Employing-supply side policies:**

- **Increased spending on training** improves productivity and reduces labour costs, or at least reduces upward pressure on labour costs.
- **Lower corporation tax**, resulting in the need to buy efficient capital equipment, can also put downward pressure on price rises.
- **Provide subsidies to troubled businesses** so they don't have to raise their prices or use subsidies to buy new capital equipment, which may result in lower long-term prices.

The effectiveness of reducing cost-push inflation:

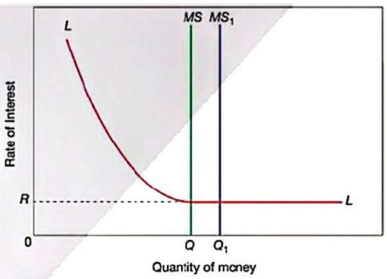
- **A rise in the exchange rate** may not mitigate inflation. If foreign firms **decide not to change** the **price of exports**, other firms, especially domestic firms, **may not respond to increased competitive pressure** to keep their cost and prices down.
- **Increased spending on training** will only be effective if their **pay stays the same**: Lower **corporation tax** may not increase investment if firms are **pessimistic about the future**.
- **Subsidies may risk increasing aggregate demand** throughout the rise in **government spending (credit creation)**, but aggregate supply will only increase if firms are **willing to cooperate and increase their efficiency**.

3.14. Keynesian and Monetarist Theoretical Approaches

- **Keynesians**: economists who think government intervention is needed to achieve full employment (level of GDP).
  - Avoidance of unemployment is a key priority.
- **Monetarists**: economists who argue that control of the money supply is essential to avoid inflation.
  - They believe an erratic change in the growth of the money supply will destabilize an already otherwise stable economy.
  - They believe the prominent role of government is to control the money supply.
  - Reducing unemployment through government spending will only raise inflation in the long run.
- **Liquidity Preference Theory**: Keynesian concept that explains why people demand money. It was showcased by the sloping curve in the graph below.
  - **Transaction Motives**: the desire to hold money for the day-to-day buying of goods and services.
    - Interest is inelastic but income elastic (influenced by income they have and can pay).
  - **Precautionary Motive**: a reason for holding money for unexpected or unforeseen events.
    - Income elastic but interest inelastic.
  - **Both are known to be active balances** (the amount of money held by households or firms for possible future usage)
  - **Speculative Motive**: a reason for holding money to make future gains from buying financial assets.

Interest inelastic

- **Known as idle balance** (the amount of money held temporarily as the returns from holding financial assets are too low)
- Government bonds are an example of hold, 500 dollars face value, with 5% fixed interest. If raised to 1,000 dollars, the interest rate will be 2.5%.
- Government bonds and interest rates **move in opposite directions** (in percentage).
- Money is held when bond prices are high and expected to fall.
- **An increase in money supply** will lead to a **decrease in interest rates**.
- Interest rates will only fall to a certain level, which will then be ineffective.



- **Liquidity Trap**: a situation where interest rates cannot be reduced to stimulate an upturn in economic activity.
- This shows the **limit of monetary policy** and how increasing the money supply will lead to a limited effect; the interest rate can only go so low; after **a certain point, it won't have an impact, and people will not bother.**

4. Government Macroeconomic Intervention

4.1. Government Macroeconomic Policy Objectives

- **A Stable Price Level**: Low and Stable Rate of Inflation.
  - The government set an inflation rate target that the central bank has to achieve and maintain.
  - A factor affecting achieving targets is the trading countries' economic state. If another country faces inflation, prices will increase for the importer country, and if it has no other importer, its production costs will rise and increase inflation pressure.
  - Another factor is rival countries; high inflation in the country's rival trading partners may also reduce pressure on the country's firms to keep their price rises low.

- **Balance of Payments Stability**
  - Maintain a balance between credit items and debt items in the long run.
    - The current account deficit creates a reduction in AD and external debt.
    - The current account surplus puts inflationary pressure and creates an opportunity cost of forgone imports.
  - Fluctuations in the current account create fluctuations in the exchange rate, destabilising and creating uncertainty.
    - Steadily raise the exchange rate to reduce inflationary pressure or reduce it to gain a competitive advantage.
- **Low Unemployment**
  - Keeps unemployment low and of short duration.
  - Reducing periods of unemployment ensures that not much income and output are lost and maintains workers' skills (becoming outdated).
- **Economic Growth**
  - Aim from economic growth or increase in economic growth rate during negative output gap.
  - Expansionary fiscal/monetary policies are used to increase AD and national output. However, it may risk inflation and a change in the current account if AD rises too much.
  - Economic growth must be sustainable so the rate of it increases in the long run.
  - Supply-side policies stimulate potential economic growth.
    - Expanding productive capacity reduces demand-pull inflation.
- **A rise in quality of life**
  - Policies address issues unrelated to increasing and maintaining the economic growth rate.
- **Sustainability**
  - Sustainable development is important to ensure the welfare of society and the economy in the long run.
  - The use of renewable energy and sources and environmentally friendly production.
- **Redistribution of income and wealth**
  - Using tax revenue of well-off individuals for welfare benefits to lower-income individuals in the economy.
  - The extent to which governments distribute income is based on different views on how significant the disincentive effect to work and enterprise of, for example, progressive taxation can be.

4.2. Links Between Macroeconomic Problems and their Interrelatedness

The Relationship Between the Internal and External Value of Money

- **Internal value is in the domestic market** and is **affected by inflation and deflation.**
- If the internal value rises above the value of a rival country, demand falls, and the external value falls. **External value is affected by depreciation and appreciation.**
- A change in **external value will also affect internal value, affecting purchasing power.**
- The exchange rate **affects the value of money** both directly and indirectly. The relationship between external and internal value is **direct.**

The Relationship Between the Balance of Payments and Inflation

- A **current account surplus** causes **short-term** upward pressure on domestic prices due to increased export revenue, **causing inflation.**
- However, due to increased current account surplus, **exchange rates rise,** and **appreciation** occurs; this leads to **reduced surplus and prices of imported goods,** which puts pressure on **domestic firms to lower prices,** thus **reducing the inflation rate.**
- **Net flow on financial accounts may reduce inflation** if multinational companies bring advanced technology and increase economic competition pressure.

The Relationship Between Growth and Inflation

- **High inflation reduces economic growth.** Decreases in net exports affect AD, lower the country's output, or reduce the output rate.
- The rise **in actual economic growth** may be accompanied by **cost-push/demand-pull inflation.**
  - **Increasing production** results in more competition for resources, including workers, which raises prices.
  - **Higher output increases income and AD,** and the gap shrinks as the economy approaches full employment.
- **Potential economic growth reduces inflation,** enabling AD to increase without pressure on price levels and reducing production costs.

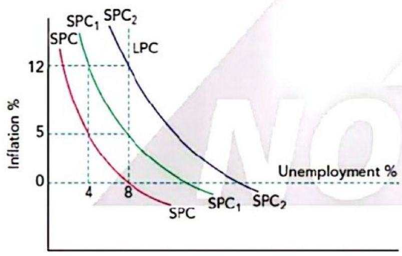
The Relationship between Growth and the Balance of Payments

- **Higher Net Exports increase GDP,** which leads to **economic growth.** This **improves the current account.**
- Potential economic growth can also have this **positive relationship** if product price and quality competitiveness **increase.**
- Possibility of an **inverse relationship in the short-run,** as imports of raw material/capital goods will increase, and the export revenue of these goods has yet to cover those import costs.
- An increase in **current account surplus increases AD,** which results in **actual economic growth.**

The Relationship between Inflation and Unemployment

- **Philips Curve**: a curve that shows the relationship between the unemployment rate and the inflation rate (money wages used as an indicator of inflation) over time.
- The government can select its best combination of unemployment and inflation and trade off the two.
- Depending on the objective, if unemployment should be reduced, expansionary fiscal/monetary policy will be applied to increase inflation and reduce unemployment.

- **Expectation-Augmented Philips Curve**: a diagram that shows that while there may be a trade-off between unemployment and inflation in the short run, there is **no trade-off in the long run.**
- This argument is by **monetarists to challenge Keynesians**.
- They argue that **policy measures used to increase aggregate demand** will **only reduce unemployment in the short run** and will only succeed in increasing **inflation in the long run.**
- Short-run Philips curves (SPC, SPC1, SPC2). AS inflation rises, unemployment falls; SPC shows us that **inflation rises from 0% to 5%**, and **unemployment falls from 8% to 4%.**
- Demand plays an important role in linking inflation and unemployment. When **demand is high, business employees need more** to meet these demands; this causes SPC1 and SPC2.
- But as **inflation rises,** businesses realise that their costs are increasing, so they will start to **cut costs, resulting in unemployment**. This is the **long-run Philip curve. At this point, no matter how high inflation gets, unemployment will not decrease.**



4.3. The Effectiveness of Fiscal Policy in Relation to Different Macroeconomic Objectives

Economic Growth and Low Unemployment Vs Balance of Payments Stability and Price Stability

- **Expansionary Fiscal Policy** increases **actual economic growth** while **reducing cyclical unemployment**; however, it can **increase demand-pull inflation** with a **deficit in the current account.**
- **Contractionary Fiscal Policy** can improve 2 macroeconomic objectives but can worsen/threaten the other two. **Lower AD helps achieve low inflation and balance the current account** but simultaneously **creates higher cyclical unemployment and lower actual economic growth.**

Crowding Out and Crowding In

- **Crowding Out Effect**: the effect of rising public sector spending driving down private sector investment.
- **New classical economists** believe that if spending by the **government is financed through increased borrowing**, that **reduces funds** available to private sectors and **raises interest rates**.
  - The high-interest rate attracts foreign financial investment, increases the exchange rate, **and reduces net exports.**
- **Crowding In Effect**: The effect of rising public sector spending leading to more private sector investment.
  - **Keynesians argue against crowding out** but agree that **crowding in occurs.** Higher government **spending leads to a multiple rise in GDP**. Higher income increases saving increases lending funds, and **encourages consumption**.

Unexpected Responses

- Uncertainty about how households and firms react **depends on their optimism about the future.**

Time Lags

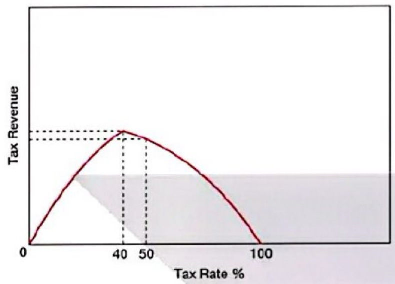
- Risks the policy **reinforcing the business cycle** rather than countering it.
- Governments must **recognise the problem** and determine the most effective policy route. Add to it the time that will be taken for **firms and consumers to react** to the changes. When that point happens, the **economy could have risen or downturned again.**

Redistribution of Income and Development versus the Incentive to Work

- Fiscal policy's use of **taxes and government spending** (transfer payments) helps influence income redistribution.
- Government spending also benefits **people by providing healthcare and education services** that help the low-income.
- However, the risk of **reduced incentives to work due to benefits and transfer payments worsens income conditions** in the future, e.g., the unemployed may experience greater poverty in the long run than if they had accepted a low-paid job now.

The Laffer Curve

- **The Laffer Curve**: a curve showing tax revenue rising at first as the tax rate increases and then falling beyond a certain point. Tax rates become counterproductive at this point.
- Apart from suggesting where **tax revenue is maximised**, the curve may show how the **same tax revenue may be received at two different tax rates**.



- At **0%**, there is no tax rate, which means **no tax revenue**.
- At **100%**, there is no tax revenue due to **no incentive work**.
- The curve suggests that tax revenue and tax rate have a **positive direct relationship** up and **until the optimal rate**, which, after that, **raises the tax rate** and discourages work and tax evasion, therefore **decreasing tax revenue**.
- The example above shows us that a cut in the tax rate from 50% to 40% will increase tax revenue.
- Economists debate the validity and usefulness of the Laffer curve, as the shape of the **curve may change according to country and time.**

4.4. The Effectiveness of Monetary Policy in Relation to Different Macroeconomic Objectives

- **Economic growth and low unemployment versus balance of payments stability and price stability**
  - The expansionary monetary policy promotes economic growth and reduces unemployment.
  - However, it may lead to demand-pull inflation and increase the current account deficit.
- **Time lag**
  - Monetary policy tends to have a shorter time lag than fiscal policy. This is because interest rates are quicker to change than tax rates and government spending.
- **How commercial banks respond**
  - Monterey policy has the additional influence of Commercial bank response.
  - Central banks may raise interest rates. However, Commercial banks (who borrow from central banks) may not pass those interest rates on to customers if they **believe high lending will generate more profit.**
  - No guarantee quantitive easing may work; an increase of liquid assets to Commercial banks through the buying of central banks may not increase Commercial banks are **pessimistic about the future and worried about the ability of borrowers to repay loans.**
- **Liquidity trap**
  - Quantitive easing is used when the interest rate is low, but it may not have an effect if it **is already low**; this creates a liquidity trap.
- **Influence of changes in other countries**
  - The trading country's central bank's response to changes may influence the effectiveness of the policy.
  - For example, if a central bank increases interest rates to discourage borrowing, but the other central bank decreases interest rates, firms will borrow from abroad instead.
- **Unexpected responses**
  - Households' and firms' responses, according to their optimism, affect the effectiveness of the policies.
  - For example, a **high-interest rate would not be discouraging if high optimism** exists.
- **Demand and supply-side shocks**
  - The assessment of macroeconomic performance and future private sector economic activity is important to the success of policies; that's why the ability to forecast is important in preventing time lags.
  - However, future forecasts are based on assumptions, not facts, and any shocks may cause adverse effects on the policies applied.
  - For example, The central bank forecasts an economic boom in a year, raising interest rates in anticipation. However, due to a global recession during the year, the interest rate increase will worsen the recession.
- **Mobility of financial investment**
  - The mobility of financial assets has made it hard to operate at an interest rate that is too different from rival countries.
  - A significant reduction in interest rates compared to rival countries may drive out hot money flows as foreign investors jump to rival country's higher interest rates. Decreasing investment and aggregate supply in the economy.
  - A rise in interest rate discourages foreign direct investment (raises foreign firm costs) and reduces demand in the country.
- **Co-ordination**
  - Fiscal and monetary policies must be coordinated and worked together.
  - Increasing income tax and interest rates will not lead to economic growth.

4.5. The Effectiveness of Supply-side Policy in Relation to Different Macroeconomic Objectives

Market-Based Supply-side Policy

- These are tools to **increase the role of market forces**:

- Cuts in direct tax
- Cuts in unemployment benefit
- Privatisation
- Deregulation
- Labour market reforms
- These policies increase economic growth and reduce inflation, but they **conflict with income redistribution objectives and lower unemployment**. This means it could be **ineffective and result in reduced economic growth** rather than an increase.
- Cuts in direct tax and unemployment benefits are used to **raise aggregate supply by increasing incentives.**
  - However, they are likely to transfer money from low-income to high-income rather than the opposite.
  - Increase gap between employment and transfer payment, but not increase employment and growth (lack of job vacancies and skills/qualification).
  - A cut in tax can have the **unintended consequence** of workers taking this opportunity to **increase leisure time rather than incentive to work.**
- Privatisation, deregulation, and labour market reforms are used to **raise aggregate supply by increasing competition and reducing the cost and effort** (for firms to meet government rules and regulations).
  - Privatisation leads to firms **charging higher product prices**, taking many **low-income consumers.**
  - Labour market reforms (reduced trade union power) and deregulation (removing minimum wage) also **reduce the relative pay of low-income consumers.**
  - Does not guarantee increased competition; privatised firms could be used to merge with other private sector firms, creating a monopoly. The reduced rules and regulations reinforce this.
  - Reducing Rules and regulations may allow firms to exploit workers, reducing labour productivity and economic growth.

Interventionist Supply-side Policy

- Tools that **increase government involvement in the economy**:
  - Government spending on education, training, infrastructure, and support for technological improvement.
- If successful, spending on Education and training can **improve all the macroeconomic objectives.** It, however, is **long-term** and **risks the demand for the skill and qualification to be outdated.**
- Improvements in infrastructure and technology help governments achieve set macroeconomic objectives. However, it risks pollution, reduced development, and some structural unemployment.

4.6. The Effectiveness of International Trade Policy in Relation to Different Macroeconomic Objectives

- Removing restrictions on imports and exports promotes free trade. Increases economic growth due to increase of comparative advantage opportunities.
  - It will be more successful if other governments follow the same decision.
- Favouring trade protectionism is more sustainable as it helps protect employment, build up infant industries and discourage dumping.
  - Trade wars could occur between other governments and members of a trade bloc, which may prevent the government from imposing trade restrictions on other members.

4.7. The Problems Arising from Conflicts between Policy Objectives

Low unemployment and economic growth versus price stability

- Expansionary fiscal/monetary policies are used to reduce unemployment and/or increase the economic growth rate but may **increase the inflation rate.**
- Conversely, Contractionary fiscal/monetary policies reduce inflation but increase unemployment and decrease economic growth.
- The introduction of supply-side policies reduces these conflicts as it works to reduce inflation and unemployment rates and increase economic growth rates in the long run.

Low inflation versus balance of payments stability

- Low inflation conflicts with a reduction in the current account deficit.
- However, a rise in interest rates (to reduce inflation) reduces investment and increases unemployment. It also appreciates the exchange rate, which may worsen the current account but might attract hot money flows.

Redistribution of income versus economic growth

- The conflict between creating greater income equality and stimulating economic growth
- Progressive income tax and increasing benefits to low-income groups may discourage work incentives and reduce foreign direct investment.
- However, the possibility of raising living standards improves the labour force and productivity.
- Foreign direct investment may increase despite a high tax rate if demand for the country is increasing and workers are more skilled.

Number of policy tools

- **Timbergen's Rule**: at least one policy measure must be for every policy aim. This is an approach used to avoid conflict between objectives.
- It uses a separate policy measure for each objective. To balance any shortcomings when it comes to certain objectives.

- Combining all 3 policies (fiscal, monetary and supply-side policy tools) is used.

4.8. Government Failure in Macroeconomic Policies

- **Government Failure**: government intervention, reducing rather than increasing economic performance.

The Lack of Adequate Information

- Underestimating the size of the multiplier may increase spending too much to make a negative output gap problem a positive output gap problem.
- Inadequate information can cause governments to employ the wrong policies, especially if you take into account future forecasting (to account for time lag); wrong predictions can cause policies applied for prevention to actually cause it.

Time Lags

- **Implementation Lag**: the time it takes to draw up and introduce a policy tool.
- **Behavioural Lag**: the time it takes for the policy tool to influence the behaviour of households and firms.
- When policy tools affect economic behaviour, economic activity might have changed, making it an inappropriate tool. Policies should aim to be counter-cyclical.

Political Pressures

- The desire to win elections can cause politicians to employ policies to increase votes; increasing government spending on pensions increases votes but may not improve economic performance.
- Powerful pressure groups and possibilities of government corruption also negatively affect economic development and growth.
- For example, corrupt government officials steal money from the government budget, which is used to be spent on consumer welfare (improve health care and education, for example)

4.9. The Effectiveness of Exchange Rate Policy in Relation to Different Macroeconomic Objectives

Balance of Payments Stability, Economic Growth and Low Unemployment versus Price Stability

- **A lower exchange rate improves the current account** of the balance of payments, **increases economic growth and reduces unemployment**.
  - Lower export prices **increase export demand and revenue**, which **increases AD and output and creates more jobs.**
  - Higher import prices may **lower demand for imports and reduce import expenditure**, which **reduces the current account deficit.**
- However, a lower exchange rate may **cause cost-push inflation and demand-pull inflation.**
  - The price of **imported raw materials/capital goods rises**, and competitive **pressure to keep prices down reduces**, which may lead to **cost-push inflation.**
  - An increase in the **net exports will raise AD,** which may **result in demand-pull if the economy approaches full employment.**
- **Raise the exchange rate, increase export prices and reduce import prices,** potentially leading to **reduced AD** and **lower cost of production.**
  - However, it leads to an increase in **deficit in the current account, reduces economic growth and increases unemployment.**

Factors that Limit the Effectiveness of Exchange Policy

- **Price Elasticity of Demand**: if demand is inelastic, a rise in the price of exports will not cause much of a rise in export revenue.
- **Speculation**: if speculation of a rise in currency grows, depreciation policy may be ineffective as private sector currency purchases would be greater than the central bank's currency sale.
- **Objectives of other countries**: governments of other countries may object if another government seeks to lower the exchange rate if it will give them a competitive advantage for its products.
- **The availability of foreign currency** and the lack of foreign currency to purchase the currency may limit how much the exchange rate may rise.

5. International Economic Issues

5.1. Policies to Correct Disequilibrium in the Balance of Payment

Components of the Balance of Payments

The Current Account

- Within the balance of payments, a record of the trade in goods, trade in service, investment income and current transfer. Consists of:
  - **Trade in Goods:** Export: rise credit items, Import: rise debit items.
    - The trade-in goods balance is the revenue earned from exports minus expenditure on imports.

- **Trade in Services:** Same as trade in goods, also known as 'invisibles'.
- **Income:** includes profits, interest and dividends from direct investments abroad and foreign earnings on investment in the country.
- **Current Transfers:** covers payments and receipts of exchange without goods and services.

- The current account balance is the balance of the overall items above. Deficit means combined debt is greater than credit combined.

The Capital Account

- Within the balance of payments, a **record of capital transfers** and the **acquisition and disposal of non-produced, non-financial assets.**

The Financial Account

- Within the balance of payments, a record of the transfer of financial assets between the country and the rest of the world. Made up of:
  - direct investment
  - Portfolio investment
  - Other investments (short-term)
  - Reserve assets

Net Error and Omissions

- A figure is included to ensure the balance of payments and balances.

Balance of Payments Disequilibrium

- **Balance of Payments Disequilibrium**: when there is an imbalance between domestic savings (credits) and domestic investments (debits).

Causes of Current Account Deficit:

- A growing domestic economy causes a decline in exports. This leads to a deficit but possibly a rise in foreign direct investment.
  - Short-term and self-correcting
- Declining economic activity in trading partners. If countries that import experience economic problems, it may cause a deficit to its partners.
  - Short-term and self-correcting
- Structural problems cause a long-term deficit.
  - Not self-correcting

A Financial Account Deficit:

- Short-term is not a problem; it could be due to excess spending in search of high-interest rates.
- Long term, however, could be because of a loss in confidence in the country's economic prospects.

5.2. Policies to Correct Balance of Payments Disequilibrium

- **Expenditure Switching Policy:** policy measure encourage people to switch from buying fore products to buying domestically-produced products. This includes foreign households to buy exports.
- **Expenditure Dampening Policy:** policy measures designed to reduce imports and increase exports by lowering demand.

Fiscal Policy

- Fiscal policy can be used as an expenditure-switching policy by imposing a tariff or increasing an existing tariff on imports.
- As a **dampening policy,** through an increase in income tax and reduced government spending, this will put pressure on domestic firms to increase efficiency and export.
- Fiscal policy **alters the country's current account position in the short term,** but **long-term** increase in tax causes side effects of **lower demand** and a reduction in **aggregate supply.**
- Imposing tariffs can also be an issue in the **trade bloc,** as well as **provoke retaliation,** and may **reduce pressure on domestic firms to be more efficient.**

Monetary Policy

- Monetary's Policy of **reducing the growth of the money supply** can be used as both an **expenditure-dampening or switching policy measure.**
- Change in interest rate effectiveness, however, depends on the **position of the country.**
  - In an economy with a **low inflation rate and current account deficit, reducing interest rates** will make the **exchange rate low,** creating **international competitiveness but risk inflationary pressure.**
- **A high exchange rate for dampening policy measures** may **reduce import demand and inflationary pressure.** Still, it can result in a **rise in the floating exchange rate** that reverses the fall in demand for imports.
- **Difficult** to control the money supply due to Commercial banks trying to find ways to **increase lending and find a way around limits.** Trying to **control can lead to the creation of new forms of money.**
- Changing interest rates takes **time to have an effect,** as well as **consumer response** to change.
- Setting/operating exchange rates have the factor of **rival countries and foreign direct investment.**
- It is **not an effective long-term solution;** the only exception is to stop maintaining the exchange rate (floating exchange rate)

Supply-side Policy

- By making domestic products more **price competitive and domestic markets attractive** for investment. This helps reduce current accounts and financial account deficits.
- More price competitiveness can be achieved by **deregulation and privatisation, increasing education and**

training, and increasing investment in quality equipment and portfolio investment.

- Some policies, like education, have long-term effects but will be **more effective than short-term solutions.**
- However, the supply-side policy outcome is uncertain;
  - Education and training to develop skills that might **no longer be in demand** in the future.
  - Privatisation may not be effective if it **causes monopolies** and does **not prioritise external costs and benefits.**
  - Trade union power may cause **inefficiency in workers.**
  - Providing subsidies may not reflect on the product, as companies may decide **not to lower prices and keep money**, as well as subsidies may cause **retaliation from foreign countries**.

5.3. Exchange Rates

- **Exchange Rate**: the price of one currency in terms of another currency.
  - **Trade-Weighted Exchange Rate**: the price of one currency against a basket of currencies. Also known as the multinational exchange rate.
  - **Real Effective Exchange Rate**: a currency's value in terms of its purchasing power.
- The Real Rate of Exchange Rate: $= \frac{\text{Nominal exchange rate x Domestic price index}}{\text{Foreign exchange rate}}$
- May rise if an appreciation or increase in the country's relative inflation rate.

The Determination of Foreign Exchange Rates

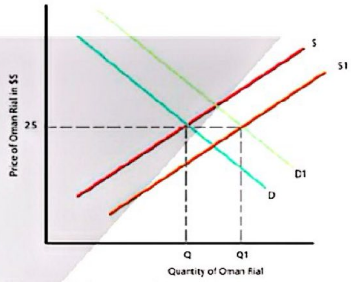
Floating Exchange Rate

- An exchange rate is determined by the supply and demand market forces.
- **Appreciation**: an increase in the international price of a currency caused by market failure.
- **Depreciation**: a decrease in the international price of a currency caused by market forces.

Advantages of floating exchange rate	Disadvantages of floating exchange rate
In theory, the exchange rate should restore the balance on the current BOP account, and a floating exchange rate helps with that.	Uncertainty as it constantly changes.
It cannot be held and used for ulterior motives like politics by the government.	Discourages trade and involvement.
Removes pressure from the government to maintain price stability using policies.	No guarantee that the floating exchange rate will fix deficit or surplus.

Fixed Exchange Rate

- An exchange rate set by the government and maintained by the central bank.
- The central bank maintains it by intervening in foreign exchange and/or changing the interest rate.
- **Hot Money Flows**: Money flows moved around the world to take advantage of changes in interest and exchange rates.
- Maintaining a fixed exchange rate is easier if that rate is set to the long-run equilibrium.
- **Revaluation**: a decision by the government to raise the international price of its currency.
- **Devaluation**: a decision by the government to lower the international price of its currency.

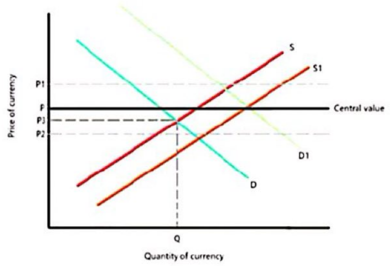


- In the graph above,1 rial = 2.5 dollars.
- If the supply shifts to S1, the central bank will buy from the currency to cause it to shift the demand curve right to D1 to maintain currency at 2.5 dollars.

Advantages of Fixed Exchange Rate	Disadvantages of fixed exchange rate
Creates certainty, which promotes international trade and investment.	The need to keep reserve, resulting in opportunity cost
It imposes discipline on the government to keep inflation low to not affect international price competitiveness and cause a downward pull on the exchange rate.	Risk of sacrificing other policy objectives to maintain a fixed exchange rate.

Managed Float

- Where the exchange rate is influenced by state intervention.
- It involves the government allowing market forces to influence the exchange rate within an upper and lower limit.



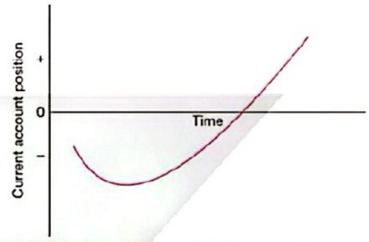
- The graph above shows a set central value of P, accompanied by an upper limit of P1 (exchange rate should not exceed) and a lower limit of P2 (exchange rate should fall beyond).
- If the exchange rate is at P3, there is no intervention by the government.
- However, if demand shifts to D1, the central bank will sell domestic currency to match the D1 with a supply shift to S1.

Marshall-Lerner Condition

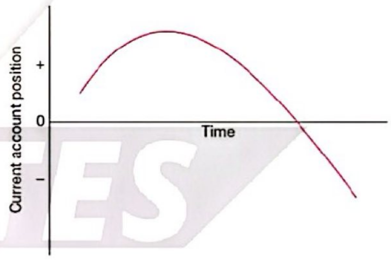
- The requirement for a fall in the exchange rate to successfully reduce a current account deficit is that the sum of the price elasticities of demand for exports and imports must be greater than 1.
- The greater the combined PED for exports and imports, the smaller the fall in the exchange rate required to improve the current account position.
- If the PED is less than 1, a revaluation of the exchange rate would be the more appropriate policy.
- Example:
  - 1 Oman rial = 2.5 US dollars.
  - 20 products in Oman would sell for 20 rials in the home country and 50 dollars in the US. **Export revenue = 400 rials (1000 dollars).**
  - 100 imports from the US, which sell at 20 dollars, are purchased by Oman at 8 rials each. **Import expenditure = 800 rials (2000 dollars).**
  - Deficit = 400 rials - 800 rials = 400 rials.
  - Then Oman rials fall in value, **1 Oman rial = 2 US dollars**.
  - The price in the US falls to 40 dollars each. Total export revenue is now **20 x 40 dollars = 800 dollars**, which is now converted to **400 Oman rials**, leaving export revenue in **Oman rials unchanged.**
    - PED for exports = 0.
  - The price of the 20-dollar product will rise to 10 Oman rials, a 50% increase. **Demand falls by 55% to 55.** Total import expenditure is now 550 rials.
    - PED for Import = -1.1
  - The deficit = 400 rials - 550 rials = 150 rials, it fell by 250 rials.

J-curve Effect

- The J-curve effect shows us that, in some cases, a fall in the exchange rate causes an increase in a current account deficit before it reduces it due to the time it takes for demand to respond.
- This is due to demand for imports and exports being **price inelastic in the short run.**
- In the long run, demand becomes more elastic, and the current account position moves to a surplus from a deficit, as shown below.



- **Marshall-Lerner condition and the J curve** work in reverse, too.
- The current account surplus will only be reduced if PED, for exports and imports, is greater than 1. A rise in the exchange rate may increase a current account surplus in the short run before reducing it in the longer run.

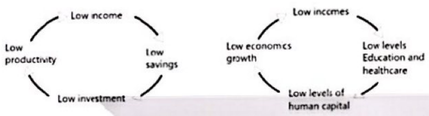


5.4. Economic Development

Classification of Economies in Terms of Their Level of Development

- **Development Economics** considers factors such as health, education, working conditions, domestic and international policies, and market conditions, focusing on improving conditions in the world's poorest countries.
- **Developed Economies**: an economy with a high GDP per head.
  - Range of well-established economic institutions
  - Advanced technological infrastructure

- Strong tertiary sector
- **Developing Economies**: an economy with low GDP per head, reliance on fewer products, a larger primary sector, and lower productivity.
  - **Poverty Cycles**: the links between, for example, low income, low saving, low investment, and low productivity.
  - **Development Traps**: restrictions on the growth of developing economies arising from low savings and investment levels.



- **Emerging Economies**: economies with rapid growth rates that provide good investment opportunities.
- The problem with classification is that, in a sense, all economies are developing or seeking to develop.
- Classification by institutions and economists is now based on the level of development (e.g., Human Development Index) economies have reached or according to income per head.

Classification of Economies in Terms of Their Level of National Income

- **GDP per capita (per head)** is calculated by dividing a nation's GDP by population.
  - It shows us, on average, the economic output per person. The higher the GDP per capita, the more well-off in the economy.
- GNI per capita is calculated by dividing a nation's GNI (Gross National Income) by population.
- The World Bank uses GNI per capita to classify every country as low-income, lower, upper, or high-income.

Threshold	GNI per capita, 2023-FY24 (Future Year) ($) \n (New)	GNI per capita, 2022-FY23 ($) \n (Previous)
Low-Income	$\leq 1,135$	$\leq 1,085$
Lower-middle Income	1,136 – 4,465	1,086 – 4,255
Upper-middle Income	4,465 – 13,845	4,265 – 13,205
High-Income	>13,845	>13,205

Indicators of Living Standards and Economic Development

Monetary Indicators

- Real GDP per capita $= \frac{\text{GDP at constant prices}}{\text{Population of country/area}}$
- GNI per capita $= \frac{\text{Dollar value of a country's final income in a year}}{\text{Its population}}$
- **NNI (Net Nation Income)** = gross national income minus the depreciation of fixed capital assets.
- **Purchasing Power Parity (PPP)**: a way of comparing international living standards by using an exchange rate based on the amount of each currency needed to buy the same basket of goods and services.
- One of the issues related to comparisons is the understated true change in output in the official real GDP, GNI, or NNI figures.
  - **Shadow Economy**: the output of goods and services not included in official national income figures.
  - It can occur due to evading tax authorities or concealment of illegal activities.
  - It can be measured using the gap between GDP and the output by measuring expenditure and income.
- Other data accuracy issues are low literacy levels, non-material goods and services (e.g., voluntary work), and measuring government spending.

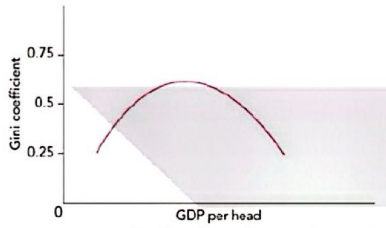
Non-Monetary Indicators

- Life expectancy, education levels, and access to healthcare.
- These help measure a country's citizens' quality of life and well-being, which is important in considering economic development.

Composite Indicators

- **Measuring Economic Welfare (MEW)**: a composite measure of living standards and factors that improve living standards.
  - It isn't easy to measure and expensive, as it involves non-market goods and services.
- **Human Development Index (HDI)**: a composite measure of living standard that includes GNI per head, education, and life expectancy.
  - HDI value shows the total data to be covered to reach 1:
    - Very high HD, high HD, medium HD, low HD.
- **Measurable Economic Welfare**: a composite measure that seeks to give a fuller of living standards by adjusting the GDP figure to consider other factors that impact living standards.
  - Factors improving living standards (including increased leisure time) are added to the GDP figure.
  - Factors reducing living standards (e.g., environmental damage) are deducted from the GDP figures.
  - In practice, putting monetary value on non-marketed goods and services is challenging.
- **Multidimensional Poverty Index (MPI)**: a composite measure of deprivation in terms of the proportion of households that lack the requirements for a reasonable standard of living.

- Six living standard indicators (cooking fuel, sanitation, safe drinking water, floor space, and assets) equal 33% of total weight.
- Two indicators of education (years of schooling and attendance) equal 33% of total weight.
- Two indicators of health (child mortality and nourishment) equal 33% of total weight.
- Households are considered multidimensionally poor if deprived of at least 33% of the weighted indicators.
- **The Kuznets Curve:** a curve that shows the relationship between economic growth and income inequality.



- As an economy develops, income becomes less evenly distributed until it reaches a certain point it becomes evenly distributed.
- This is because only a few gain high rewards as the economy develops, so there is a broader income gap. However, a middle grade emerges due to progressive tax, and more income is spent on social welfare and public goods.
- Only applicable to developing countries. It does not explain the rise in inequality in developed.
- All developing countries do not use it.

5.5. Comparisons of Economic Growth Rates and Living Standards

Overtime

- Real GDP per head is usually used as the main indicator. In a particular year, if **real GDP per head is greater than the previous year,** it is expected that the population is experiencing a **higher living standard**.
  - However, Real GDP is **not evenly distributed**. Higher GDP per head may not mean everyone is well off.
- The change in real GDP **may not reflect the true change in the number of goods and services** households can enjoy due to **undeclared economic activity over time.** For example, more weapons produced in war may not mean consumers' quality of life is improving.
- In the **long run**, producing **more capital goods will result in higher consumer products and**, thus, **higher living standards.** However, in the **short run**, the **shift of resources may cause a dip in living standards.**
  - More goods/services are not guaranteed to ~~result in better living standards~~. As **more is supplied, more will be demanded**, potentially at an even faster rate than can be supplied.
- **Real GDP measures the quantity of output, not quality.** Output may rise, but if quality falls, so does living standards.
  - In practice, however, **quality rises over time.**
- **Working conditions** also tend to **improve over time, with working hours falling.**
  - If **real GDP per head stays the same**, while working conditions rise and working hours fall, people's living standards will increase.
  - If improved working conditions over time lead to a **decline in the quality of the environment**, **lower living standards** will happen, but **GDP may rise** (more resources being devoted to cleaning up the environment)

Between Countries

- In Countries with Higher GDP per head, the citizens are likely to enjoy higher living standards than those with lower GDP per head. However, this is not always the case.
- Real GDP per head must be converted into a common currency to compare living standards.
- To avoid comparisons being distorted by exchange rate changes, economists would **adjust the exchange rate to consider their purchasing power parity.**
- Example:
  - 2.5 Oman rials = 1 US dollar.
  - **If Oman's real GDP per head was 10,000 rials, and the US GDP per head was 20,000 dollars**. This would mean when Oman's real GDP is converted to US dollars (4,000 dollars), the people in the US are **five times better off than Oman's people.**
  - However, if 1 dollar can buy more in the US than in Oman, using the exchange rate will **exaggerate Oman's output.**
  - **A basket of products may sell for 200 dollars in the USA and 1400 rials in Oman.** This would mean that in terms of ability to buy (Purchasing parities), **7 rials are worth 1 dollar.**
  - Using this as a basis to convert Oman's output to US dollars would show that the people in the **US are 14 times better off than people in Oman.**
- Even if a country has a higher real GDP than another country using PPP, it does not guarantee that the population will enjoy a higher living standard.
  - This is due to **uneven income distribution**, e.g., Kuwait has one of the highest GDP per capita, but many immigrant workers receive low wages.
- Like comparisons over time, **factors not measured in real GDP per head** must be **accounted for in assessing living standards.**

5.6. Characteristics of Countries at Different Levels of Development

Population Growth and Structure

Demographic Factors

- **Birth Rate**: the number of live births per thousand population in one year.
  - $\frac{\text{Number of live births}}{\text{The total population}} \times 1000$
  - It is also referred to as ~~**crude birth rate**~~, as it only gives basic information on births (not, for example, the mother's age).
- **Death Rate**: the number of deaths per thousand population in one year.
  - $\frac{\text{Number of death}}{\text{The total population}} \times 1000$
  - Death rates include children under 1, even tho infant mortality is measured separately.
  - **The natural increase in population** is when the birth rate exceeds the death rate.
- **Infant Mortality**: the opposite of Birth rate, the number of deaths of children under 1 per thousand of the population in one year.
- **Net Migration**: The difference between migration into and out of a country divided per thousand
  - $\frac{\text{Difference between immigration and emigration}}{\text{The total population}} \times 1000$
  - If immigration into the country is greater than emigration out, net immigration or positive net migration happens.

Causes of population changes:

- As countries develop, **population growth goes down. The death rate decreases**, but the **birth rate drops more.** This is due to a decline in infant mortality, increased cost of raising children and women's participation in the labour force, better healthcare, housing, nutrition, etc., contributing to people living longer.
- **Population growth** can still occur in these countries through **increased net immigration** (higher living standards and income attract people).
- **Low development** is associated with **high population growth rates** due to the **birth rate exceeding the death rate**.
  - The need for children to support parents, lack of availability/methods of birth control, low cost of raising kids, lack of education for women and high infant mortality (more children to compensate).
  - This results in a **low average population age** and a **high dependency ratio** (people who can't be in the workforce must be provided for).
- High development creates an **ageing population** (due to a decreasing birth rate and a decreasing death rate), and this also makes a **high dependency ratio** and **greater cost for health care and pensions.**
  - Governments have **increased** the **retirement age to reduce the cost of pensions.**

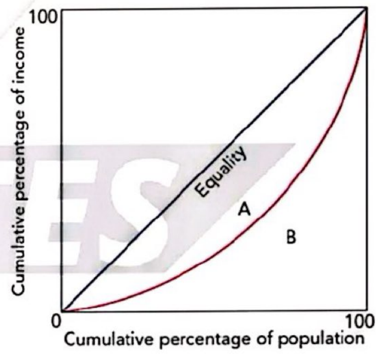
The Optimum Population

- **Optimum Population**: the size of the population that maximises GDP per head.
- As the population grows, it makes better use of the stock of capital and land. This increases returns enjoyed by population growth.
- A country is considered underpopulated if the population is below the optimum level.
- The country is considered overpopulated if the population exceeds the optimum level, and decreasing returns are happening.
- In practice, the situation is more dynamic (constantly in change). Technical knowledge is continually improving, and due to the number of other factors constantly changing, there is no fixed optimum population for a country.

Level of Urbanisation

- Countries with low income per head usually have a high proportion of the population living in rural areas.
  - Rural-urban migration rates are high in these countries, putting pressure on the urban areas' infrastructure, housing and schools.
- Most of the population in countries with high income per head live in urban areas, so they have small growth in urban population.
  - In some cases, people are migrating to rural from urban due to technological advances allowing more "work from home".

Income Distribution



- **Gini Coefficient**: a numerical measure of inequality
  - Calculated as, $\frac{\text{Area A}}{\text{Area A + Area B}}$
  - If the distribution is equal, the coefficient is 0.
  - A coefficient of 1 would mean complete inequality (one person getting all the income).
  - The higher the coefficient, the more unequal the distribution of income.
- **Lorenz Curve**: a graphical representation of inequality.
  - The 45° line shows a completely equitable distribution of income.

- The curve shows the actual distribution of income or wealth.
- Area A is the difference between the curve and the line. It represents inequality; the smaller the Area of A, the more equal the distribution.

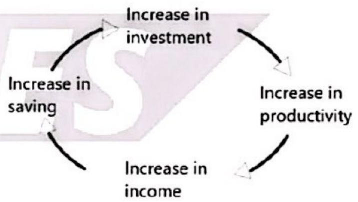
Economic Structure

- **Primary Sector**: extraction of natural resources to be used and processed into goods and services.
  - Agriculture and Extractive Industries.
- **Secondary Sector**: the manufacturing of goods and services from natural resources.
  - Manufacturing industries and construction industries.
- **Tertiary Sector**: Service industries to both consumers and firms.
- Countries with **high income per head and high development** have a large portion of their labour force in the **tertiary sector**.
  - They export mainly manufactured goods and services.
- Countries with **low income per head and development** depend more on the **primary sector**.
  - More **vulnerable to natural forces**. High rain seasons/droughts will affect agriculture-dependent economies, leading to famine, and if they are exported to almost no foreign currency earning.
  - The majority of Export revenue is generated from primary products. **Frequent and significant fluctuations in price** (through demand and supply changes) make them **vulnerable in their trading relations.**
- **Industrialisation** of the economy (due to economic development)
  will increase the relevance of the secondary sector, and this will move the labour force from the primary industry, reducing its significance.
  - Primary goods are **inelastic**, with manufactured ones being more **elastic**. **Low prices and revenue** for direct goods compared to manufactured goods make them hard to export, creating a **decline in primary goods.**
- As the economy develops, the employment of the labour force will continue shifting towards the tertiary sector. **Increases the value of output due to higher labour and capital productivity.**

5.7. Relationships Between Countries at Different Levels of Development

International Aid

- **Foreign Aid**: assistance given to developing economies on favourable terms.
- **Tied Aid** comes with conditions
- **Untied Aid** is given without conditions
- **Bilateral Aid** is given to one country by another.
- **Multilateral Aid** is given by countries to international organisations, who distribute it to other countries.
- **Voluntary Aid** incorporates volunteers who donate their labour as a form of aid.
- **Official Development Assistance (ODA)**: government aid that promotes and specifically targets developing countries' economic development and welfare.
- **Reasons for Aid**:
  - **Tied, bilateral aid may be given to promote the industries** of the donor country through sending aid and expecting the recipient country to invest/spend on products from the donor country, restricting its import. **Increases donor countries' exports.**
  - **Untied aid may promote the donor country's exports** through better economic growth in the recipient country, as they may **import more** due to a **good relationship**.
  - **Bilateral aid** may be used to gain **political influence.** The recipient government may be **pressured or inclined to support the donor government** in its disputes with other government(s).
  - Both **bilateral and multilateral aid** may be given to **influence the economic policies** of the recipient government. E.g. Aid given on the condition ends the use of child labour.
  - Bilateral and multilateral aid can be given for humanitarian motives, such as crisis aid (to save lives during natural disasters and famines). The development of other countries increases global GDP and international trade and reduces the risk of negative external shock.
- **The effects and importance of aid:**
  - Aid helps the recipient country **experience increases in its income per head and development** by providing investment/finance to stimulate them.
  - If investment is encouraged, a virtuous cycle may be created.



  - **Dependency**: a situation where the economic development of the recipient economy is hindered by its relationships with donor economies.
  - Recipients may become indebted, with high-interest payments, for past loans to high-income countries; they may be paying more than the aid received.
  - Aid in terms of advice and policies suggested/imposed may not always be suitable towards the recipient country.

Trade and Investment

- Low-income/some middle-income countries **favour trade more than aid.** International **trade has many benefits compared to aid**:
  - **Economies of scale** are becoming possible because of the larger market.
  - **Increased competition** encourages domestic entrepreneurs to **innovate** and look for **new production techniques**.
  - Trade leads to the **transfer of skills and technology.**
  - Specialisation and trade **increase income** and provide **increased savings** that can be used for **investment.**
  - Expansions caused by trade **increase employment.** expand **spending power** and **increase demand** for domestic output.
- Low-income/middle-income countries specialise in agricultural products, this puts them at a disadvantage due to:
  - The income elasticity of demand for primary goods is very **inelastic**, causing **little extra demand with increased world income.**
  - Producers of manufactured goods in high-income economies have some **monopoly power**, which helps maintain their high prices. Puts primary goods at a disadvantage in trade.
  - **Subsidies** provided in the USA and Europe put downward pressure on global agricultural prices. **Unfair competitive advantage** due to subsidies and a **deflated global price.**
- Countries adopt **import substitution policies** (less import of manufactured to develop their own manufacturing industries) and **attempt to diversify their economies** in attempt to combat these exploitative trading patterns.
- **Export-led growth** through industrialisation gives them a more comparative advantage in industries previously dominated by high-income economies.
- Investment flows between countries in search of profits, interest and dividends.
- Many low-income/middle-income countries incur a **deficit in their current account**, which makes investment impossible unless a **surplus in the financial** account covers it. This makes Low income/middle income countries seek to **attract direct and portfolios from other countries.**

Role of Multinational Companies (MNCs)

- **Multinational Companies (MNCs)**: a business with a parent company based in one country but with production or service operations in at least one other country.
- **The Beneficial activities of MNCs:**
  - Bring in new technology and new ideas into host countries.
  - Add to GDP and exports, contributing to the economic welfare of the host country.
  - Generate employment opportunities
  - Led to an increase in Foreign direct investment (FDI)
- **Negative consequences of MNCs:**
  - It may **drive domestic firms out of the market** (especially infant and declining industries), but this does **not create higher employment.**
  - **Depletion of non-renewable resources** and the creation of **pollution.**
  - May send **profits back to home country** and **employ foreign** rather than domestic workers.
  - Depending on MNCs' power, it may **pressure the government** into employing policies that benefit them but not the economy and negotiate favourable tax breaks and exemptions from some environmental laws.
  - Monopoly Power may be developed, which gives them the **power to drive down prices of raw material supplies** in the host country.

Foreign Direct Investment (FDI)

- **Foreign Direct Investment (FDI)**: setting up production units or purchasing existing production units in other countries. It is not the same as a portfolio investment (purchasing shares in another country).
- Low-income/middle-income countries benefit from FDI through MNCs, which helps the country's infrastructure.
- **Large inward flows of FDI** indicate that a country is expected to grow rapidly and provide large markets for MNCs or markets with low production costs or an **abundant supply of raw materials.**
- **Measures to attract FDI:**
  - Low corporation tax
  - Good education system
  - Few rules and regulations for firms.
  - Government subsidies

External Debt

- **External debt includes:**
  - Loans have not been repaid, and interest payments on loans have not been made to foreign banks, governments and international organisations.
- **Causes of Debt:**
  - Structural current account deficit
  - Overconfidence in the value of loans that could be repaid.
  - Inefficient use of borrowed funds
  - Unexpected events
- **Consequences of Debt:**
  - The opportunity cost of repaying debt with the welfare of the population.
  - A high level of debt may make it difficult for low-income/middle-income countries to attract more development funds.

Role of the International Monetary Fund (IMF)

- **The International Monetary Fund (IMF)** works to achieve sustainable growth and prosperity for its 190 member countries.

- **The primary aim of IMF:**
  - To promote international monetary cooperation
  - To facilitate the expansion and balanced growth of international trade
  - To promote exchange rate stability
  - To assist in setting up a multilateral system of payments
  - To make resources available (with adequate safeguards) to members experiencing balance of payments difficulties.
- These activities create a stable system of international payments and exchange rates necessary for trade between 2 countries.
- **Functions of IMF:**
  - **Surveillance and technical assistance**, which encourages countries to adopt sound economic policies, helps promote global growth and economic stability.
  - **Lending** is used to help members experiencing difficulties in financing their balance of payments.

Role of World Bank

- The World Bank is an international development organisation owned by 187 countries.
- The World Bank supports low/middle-income countries through **internal investment payments** such as building new roads, improving infrastructure and constructing new health facilities.
- **The International Bank for Reconstruction and Development** (IBRD) assists middle-income and creditworthy poorer countries, while the **International Development Association** (IDA) focuses on the poorest economies, which are given grants.
- **They provide loans that cover the following:**
  - **Health and education**
  - **Agriculture and rural development**
  - **Environmental protection**
  - **Infrastructure**
  - **Governance (anti-corruption reasons)**
- These loans are given to recipients who want to create wider-reaching changes to their economic policies.

5.8. Globalisation

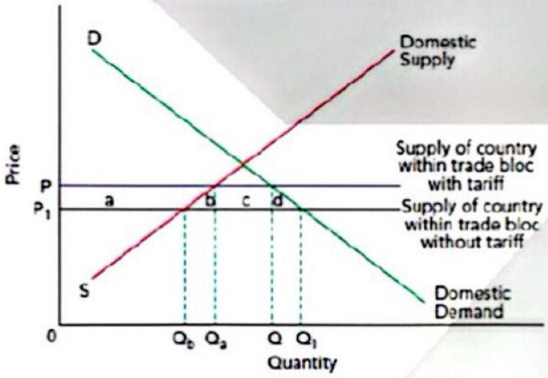
- **Globalisation:** increasing interdependence of world economies due to the growing scale of cross-border trade of commodities and services, the flow of international capital and the vast and rapid spread of technologies.
- **Causes of globalisation:**
  - **Reduction in trade barriers** - makes product, capital and labour markets easily accessible for different countries.
  - **Improved telecommunications** - more accessible to keep in contact with production plants, making coordinating the production process easier. The increase in online stores makes consumption more convenient.
  - **Reduced supply chain costs** - improved transport, as made resource mobility easier and cheaper.
  - **Deregulation of financial markets** - more freedom for banks to use and allocate their capital, increase in foreign direct investment, portfolio and direct investment.
  - **Technological change** - technological advancement has made many processes more efficient and effective.
- **Consequences of Globalisation:**
  - **Improved standards**: Increases comparative advantage, allows for increased GDP and economic growth and increased living standards.
  - **Increase of Foreign Direct Investment**: Reduced trade restrictions attract MNCs, which increases FDI.
  - **Greater Variety of Products**: removal of trade restriction encourages imports.
  - **Structural Unemployment**: Industries expand due to greater international competition while some decline.
  - **Interdependence of Economies**: the increased possibility of demand and supply side shocks; these may be adverse shocks like natural disasters.
  - **International Investment**: free flow of portfolio and direct investment runs the risk of a negative multiplier effect if economic conditions worsen and the investment is drawn out quickly.
  - **International Coordination of Policies:** to prevent policy conflicts and improve the effectiveness of policies, but not very easy to achieve.

Economic Integration

- **Economic Integration**: an arrangement among nations that typically includes reducing or eliminating trade barriers and coordinating monetary and fiscal policies.
- **Trade Blocs**: a regional group of countries that have entered trade agreements.
- **Free Trade Areas:** a trade bloc where member governments agree to remove trade restrictions among themselves and determine their external trade restrictions/policies towards non-members. E.g. NAFTA.
- **Custom Unions**: a trade bloc with free trade between member countries and a common tariff on imports for non-members. E.g. SACU
- **Monetary Union**: a trade bloc with free trade between member countries, a common external tariff, the same currency, and the same monetary and exchange rate policies. E.g. EU
- **Economic Unions**: a trade bloc with free trade between member countries, a common external tariff, economic policies, and a common currency. E.g. 7, the original states joining together into the UAE or the 13 states that formed the USA.
  - The full economic union is the final stage of integration.

Trade Creation and Trade Diversion

- **Trade Creation**: where high-cost domestic production is replaced by more efficiently produced imports within the customs union.
- Before joining the customs union, the price was P and quantity Q (Qa supplied by domestic producers, Qa- Q is the amount imported).
- After joining the customs union, it can import without tariff, which pushes the price to P1 and increases quantity to Q1.
- Here, consumers gain low prices and high quantity, while firms lose sales and lower prices. This could be because of a resource shift to specialise in price-competitive goods.
- Lower price increases consumer surplus by a, b, c, d amount.
- Producer surplus falls by an amount and tariff revenue by c, giving a net gain of b and d.



- **Trade Diversion**: where trade with low-cost countries outside a customs union is influenced by higher-cost products supplied within.
- Initially, the country bought from the most efficient country and placed tariffs on imports, price at P and quantity at Q, with tax being c + e.
- The trade is switched to another member country when joining the customs union. Price falls to P1, and quantity rises to Q1.
- Consumer surplus increases by a, b, c, d amount.
- If producer surplus falls by an amount, tariffs will decrease by c+ e.
- If areas b and d are less than area e, welfare will be reduced.